Influence of Games on Cooperative Behaviours: A Multilevel Meta-Analysis

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Abstract

Objective. This meta-analysis examines the effects of cooperative and prosocial video game play on prosocial behavior, drawing on both game content (prosocial vs. neutral/violent) and game context (cooperative vs. competitive/solo play). **Methods**. A systematic literature search identified 45 eligible studies (88 effect sizes) published between 2014 and 2023. A multilevel random-effects meta-analysis was conducted using the metafor package in R, accounting for the nested structure of multiple effect sizes within studies. Effect sizes were sign-corrected based on construct interpretation coded with the aid of a large language model (LLM). Publication bias was assessed via funnel plot inspection, Egger's regression test, and rank-order correlation. **Results**. The multilevel random-effects model yielded an overall effect of $d = 0.419$ (SE = 0.091), 95% CI [0.241, 0.598], $z = 4.6$, $p < .001$, indicating a positive small-to-medium effect of prosocial and cooperative game play on prosocial outcomes. Heterogeneity was substantial ($I^2_{\text{total}} = 85.5\%$). Moderator analyses revealed that neither game content (prosocial vs. neutral/violent: $b = -0.50$, $p = .110$) nor game context (cooperative vs. competitive/solo: $b = -0.15$, $p = .632$) significantly moderated the effect. Egger's test indicated funnel plot asymmetry consistent with publication bias ($z = 4.72$, $p < .001$). **Conclusions**. Cooperative and prosocial game play is associated with positive social outcomes of moderate magnitude. The high heterogeneity and evidence of publication bias suggest that the true effect may be somewhat smaller. Future research should use pre-registered designs and standardized outcome measures to strengthen causal inferences.

Keywords: cooperative games, prosocial behaviour, meta-analysis, video games, helping behavior

Influence of Games on Cooperative Behaviours: A Multilevel Meta-Analysis

The video games industry has grown significantly over the last few decades. According to Newzoo (2023), the global games market was worth over $180 billion in 2023, and there are approximately 3.3 billion gamers worldwide. Video games are popular, diverse, easily accessible, and played by people of all ages and nationalities. Accordingly, research on the influence of games on human behavior, cognition, and affect has grown rapidly.

Initially, researchers focused mainly on the antisocial effects of playing video games. Anderson et al. (2010) published a comprehensive meta-analysis on the effects of playing violent video games, demonstrating that exposure to violent content constitutes a causal risk factor for aggressive behavior, aggressive cognition, and aggressive affect, while being negatively associated with empathy and prosocial behavior. The first meta-analysis that examined the prosocial effects of playing video games was published by Greitemeyer and Mügge (2014). Analyzing data from 98 independent studies with over 36,000 participants, they demonstrated that violent video games increase aggression-related variables and decrease prosocial outcomes, while prosocial video games have the opposite effects.

These early studies focused primarily on the *content* of video games—whether the game portrayed prosocial or antisocial interactions. The dominant theoretical framework employed was the General Learning Model (GLM; @Buckley2006), an extension of the General Aggression Model. According to the GLM, interacting with video game content can have both short-term and long-term effects on players: games provide behavioral models, reinforce certain actions, and prime prosocial or antisocial cognitive scripts, gradually shaping attitudes, schemas, and affective traits through repeated exposure.

However, this content-focused approach overlooks one vital aspect: the *context* in which a game is played. A growing body of research indicates that playing the same game cooperatively, alone, or competitively produces meaningfully different effects on the player. For example, Jerabeck and Ferguson (2013) examined how game content and context jointly influence aggressive and prosocial behavior, finding that cooperative play was associated with less

aggressive behavior regardless of whether the game was violent or not. Shoshani and Krauskopf (2021) further demonstrated, in a large sample of children (N = 845), that playing a violent multiplayer game cooperatively (vs. solo) resulted in greater prosocial behavior, with psychological need satisfaction—competence, relatedness, and positive emotions—acting as a mediator. These results suggest that the social context and the psychological experience of playing may matter more than the formal content of the game.

More recent studies have extended these findings. Zhang et al. [-@Zhang2025] found in an experimental study with preschoolers that cooperative (vs. competitive) game play increased sharing behavior, though with notable gender moderation. Sherrick et al. [-@Sherrick2025] showed that cooperative game settings promote prosocial injunctive norms among multiplayer online players. At the neural level, there is converging evidence that empathy-training games can produce functional changes in emotion-regulation circuits [@Kral2018].

Despite growing evidence, the field still lacks a comprehensive meta-analytic synthesis that simultaneously accounts for both game *content* (prosocial vs. neutral or violent) and *context* (cooperative vs. competitive or solo), while adequately controlling for the nested structure of effect sizes within studies. The present meta-analysis addresses this gap.

Current Study

According to Batson [-@Batson2003, p. 463], prosocial behaviors refer to “a broad range of actions intended to benefit one or more people other than oneself—behaviors such as helping, comforting, sharing and cooperation.”. We distinguish two types of game-based pathways to prosocial outcomes. First, games may be prosocial by *design*: their mechanics, narrative, and goals are coherently oriented toward promoting prosocial change. Second, many commercial games have prosocial effects without that being their primary intention—they contain prosocial *content* (e.g., mechanics, story, role) or are played in a prosocial *context* (i.e., cooperatively).

A further complication is that the boundary between prosocial, neutral, and antisocial games is not sharp. For instance, some games require cooperating with teammates to fight against opposing teams, while others involve violence in the service of saving the world. Rather than

imposing an a priori classification, we follow the authors of original studies in their characterization of the game conditions as prosocial or cooperative.

In summary, this meta-analysis pursues two primary objectives: (1) to estimate the overall effect of prosocial and cooperative game play on social outcomes, and (2) to examine whether game content and game context moderate this effect. Given previous findings, we hypothesize a positive overall effect ($d > 0$) and expect the effect to be larger for cooperative game contexts than for prosocial content alone.

Methods

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study.

Literature Search and Study Selection

In search of relevant studies, we used Google Scholar as the primary database. The following search string was used:

> *"prosocial games" OR "cooperative games" AND "prosocial behavior" OR "cooperative behavior" OR "prosocial attitude" OR "prosocial behaviour" OR "cooperative behaviour" –equilibrium*

The search was conducted on 22 February 2021 and updated in 2023 and 2025. In the first stage, we analyzed 1,407 abstracts to identify 85 studies that appeared suitable. Upon further reading, 45 studies were selected as relevant and providing sufficient data for meta-analytic inclusion.

Inclusion criteria: (1) published between 2014 and 2025; (2) written in English; (3) included either a cooperative or prosocial game as a key variable; (4) measured at least one social outcome variable (prosocial behavior, attitude, or affect); (5) reported sufficient statistics to compute or convert an effect size.

Exclusion criteria: Studies were excluded if they were qualitative, did not include a control or comparison condition, or if the statistical information was insufficient for effect-size

computation.

A PRISMA flow diagram summarizing the study selection process is presented in **?@fig-prisma**.

```
# [PRISMA diagram do wygenerowania lub wklejenia jako obraz]
```

Coding Procedure

For each included study, two independent coders extracted the following information: citation key, study design (experimental vs. correlational), participant age range, game name, game type (video vs. non-video), whether the game was prosocial by content (`prosocial`), cooperative by context (`cooperative`), and multiplayer. Inter-rater agreement was assessed with Cohen's κ.

```
# [IRR - Cohen's kappa do obliczenia na bazie podwójnego kodowania]
```

LLM-Assisted Coding of Effect Direction

A central challenge in meta-analyzing this literature is that measures of prosocial outcomes differ in their scaling direction: for some constructs (e.g., helping behavior, empathy, sharing), higher scores indicate more prosocial responding; for others (e.g., aggression, hostility, competition), higher scores indicate less prosocial responding and the scale must be reversed. To address this systematically, we used a large language model (LLM) to read the description of each measured construct and assign a direction code:

- `+`: A higher value on this measure reflects a more prosocial outcome; the raw effect size should be retained as is.
- `-`: A higher value reflects a *less* prosocial (or more antisocial) outcome; the sign of the effect size should be reversed.
- `?`: Ambiguous; requires manual verification.

Two LLMs were used in parallel to code the direction of each effect size: ChatGPT (GPT-4o, OpenAI, December 2025) and Perplexity AI (default model, December 2025). Out of

88 effect sizes, ChatGPT assigned 70 as positive direction, 13 as requiring sign reversal, and 5 as ambiguous. Ambiguous cases were resolved through consensus among the research team.

Inter-rater agreement between the two LLMs was computed using Cohen's kappa. For the full set of 88 effect sizes, $\kappa = 0.63$ ($z = 7.65$, $p < .001$), indicating substantial agreement. For the 75 cases where neither model returned an ambiguous code, $\kappa = 1$, indicating perfect agreement.

This approach is consistent with emerging validation work showing that LLMs can successfully approximate the direction of effect in meta-analytic coding tasks with high concordance rates (Blaizot et al., 2022).

Effect Size Computation

All effect sizes were converted to Cohen's *d* (standardized mean difference, SMD). The conversion depended on the type of statistic reported in the original study and the structure of the comparison (see **?@tbl-conversions** for the formulas used).

Table 1

Effect size conversion formulas by source statistic type.

Source statistic	Formula / Package
Means and SDs (two groups)	metaConvert::es_from_means_sd()
Means and SDs (within)	metaConvert::es_from_means_sd(), n=n
t-test (two groups)	esc::esc_t()
t-test (equal groups)	esc::esc_t(totaln=n)
F-test	esc::esc_f()
Chi-squared	esc::esc_chisq() / esc::esc_rpb(phi)
z-statistic	effectsize::z_to_d()
Point-biserial r or β	esc::esc_rpb()
Regression coefficient b	esc::esc_B(sdy=sd_Y)

The variance of each SMD was computed using the standard formulas appropriate for the

type of statistic (e.g., $v = \frac{n_1+n_2}{n_1 n_2} + \frac{d^2}{2n}$ for two-group designs). After conversion, each effect size was sign-corrected using the `direction_sign` variable derived from LLM coding, such that a positive d always indicates an effect in the prosocial direction.

Statistical Analysis

To account for the nested structure of the data—multiple effect sizes from the same study—we employed a **multilevel (three-level) random-effects meta-analysis** using the `metafor` package [@Viechtbauer2010] in R. The model included two random-effects components:

1. τ^2_{within}: variance of effect sizes *within* studies (e.g., different outcome measures, groups, or time points within the same article).
2. τ^2_{between}: variance *between* studies (differences across articles as whole units).

The model was fitted with a known within-study covariance matrix $\mathbf{V}$, where off-diagonal elements were estimated assuming a correlation of $\rho = 0.60$ between effects from the same study (following Cheung, 2019). Sensitivity analyses were conducted with $\rho \in \{0.40, 0.80\}$.

Heterogeneity was quantified using two I^2 statistics (Viechtbauer, 2021):

$$I^2_{\text{within}} = \frac{\tau^2_{\text{within}}}{\tau^2_{\text{within}} + \tau^2_{\text{between}} + \sigma^2}$$

$$I^2_{\text{between}} = \frac{\tau^2_{\text{between}}}{\tau^2_{\text{within}} + \tau^2_{\text{between}} + \sigma^2}$$

where σ^2 is the average sampling variance across effect sizes.

Moderator analyses were conducted as meta-regressions within the same multilevel framework, examining:

- Game type: `prosocial` (content-based) vs. `cooperative` (context-based)
- Study design: experimental vs. correlational
- Game medium: video game vs. other (e.g., board game, non-digital)
- Participant age group

Publication bias was assessed using: (1) visual inspection of a funnel plot, (2) Egger's weighted regression test (regtest), and (3) Kendall's rank-order correlation (ranktest). Influential observations were identified using Cook's distance and DFFITS statistics.

All analyses were conducted in R (version ≥ 4.3) with packages `metafor`, `esc`, `metaConvert`, `effectsize`, `tidyverse`, and `knitr`.

Results

The final sample comprised 88 effect sizes nested within 47 independent studies, published between 2014 and 2023. The total number of participants across studies was N = 12,768. Of the included effect sizes, 86.4% came from experimental studies and 13.6% from correlational designs. 84.1% of effect sizes involved video games; the remainder came from non-digital game studies (e.g., board games, structured cooperative activities). 52.3% of effect sizes involved a game that was prosocial by content, while 58.1% involved a cooperative game context.

Table 2 provides an overview of included studies.

Convergent Validity of Effect Size Computation

To validate the effect-size computation pipeline, we correlated the newly computed SMDs (`SMD_R`) with SMDs provided directly in the original dataset (`SMD`). The two indices were strongly correlated, $r(81) = 0.95$, 95% CI [0.93, 0.97], indicating high convergent validity of the conversion procedure.

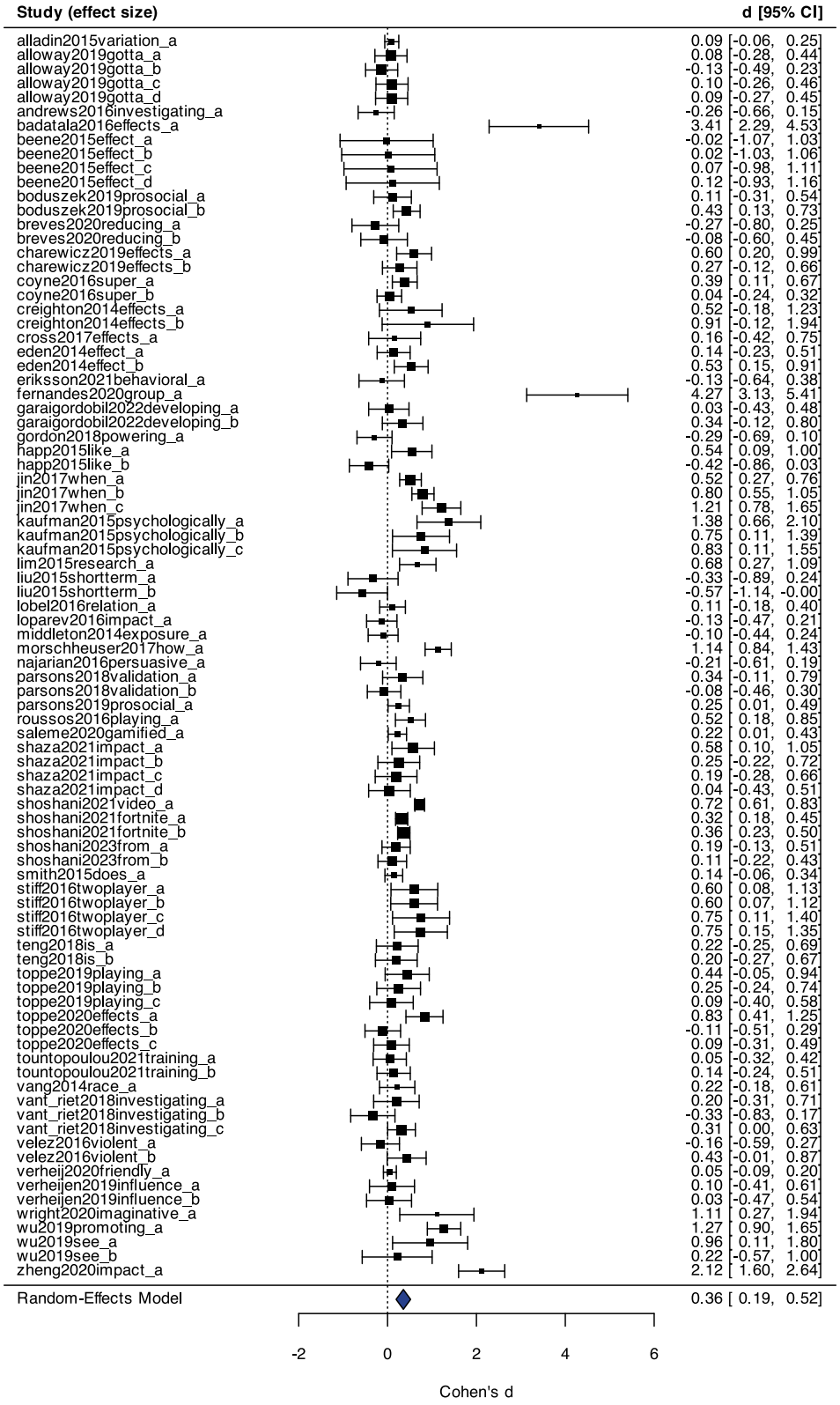
Overall Meta-Analytic Effect

The multilevel random-effects model yielded an overall estimated effect of d = 0.356 (SE = 0.083), 95% CI [0.192, 0.519], $z = 4.26$, $p < .001$. This indicates a positive and small-to-medium effect of prosocial and cooperative game play on prosocial outcomes.

Heterogeneity was substantial: $I^2_{\text{between}} = 63.3\%$, $I^2_{\text{within}} = 17.2\%$, total $I^2 = 80.5\%$. The between-study variance component was $\hat{\tau}^2_{\text{between}} = 0.2358$ and the within-study variance component was $\hat{\tau}^2_{\text{within}} = 0.064$. These results indicate that a larger proportion of heterogeneity was attributable to differences between studies than to within-study variance.

Figure 1

Forest plot of included effect sizes (multilevel model). Diamond indicates the pooled estimate.



Sensitivity Analyses: Correlation Assumption

Table 6

Sensitivity analyses: pooled effect across assumed within-study correlations (ρ).

ρ	d	SE	95% CI	z	p	I^2
0.4	0.356	0.083	[0.192, 0.519]	4.27	1.97e-05	80.5
0.6	0.356	0.083	[0.192, 0.519]	4.26	2.01e-05	80.5
0.8	0.355	0.083	[0.192, 0.519]	4.26	2.04e-05	80.4

The main findings remained robust across sensitivity analyses with assumed within-study correlations ranging from ρ = 0.40 to 0.80. The pooled effect estimates were comparable, indicating that inferences about the overall effect of prosocial and cooperative games on social outcomes are not unduly sensitive to the choice of within-study correlation structure.

Sensitivity Analyses: Large-Effect Exclusion

Table 7

Three largest effects (selected for sensitivity analysis).

cite	lab	SMD_signed	abs_d
fernandes2020group_a	Fernandes_2020	4.269	4.269
badatala2016effects_a	Badatala_2016	3.406	3.406
zheng2020impact_a	Zheng_2020	2.118	2.118

Table 8

Sensitivity analysis: full sample vs model excluding the 3 largest effects.

Model	k effects	d	SE	95% CI	p	I^2 total
Full sample	88	0.356	0.083	[0.192, 0.519]	< .001	80.5
Without 3 largest |d| effects	85	0.245	0.049	[0.148, 0.342]	< .001	58.1

To ensure transparent reporting, we present both the full-sample model and a sensitivity model excluding the three largest effects (identified a priori by absolute magnitude of d among effects with $|d| > 2$). The substantive conclusions are interpreted based on the full model, while the reduced model is treated as a robustness check.

Heterogeneity and Influence Diagnostics

Egger's weighted regression test indicated significant asymmetry in the funnel plot ($z = 3.36$, $p < .001$). Kendall's rank-order correlation between effect sizes and their precision was $\tau = 0.156$ ($p = 0.031$), suggesting publication bias.

Figure 2

Funnel plot of standard error by Cohen's d. Vertical line indicates the pooled estimate.

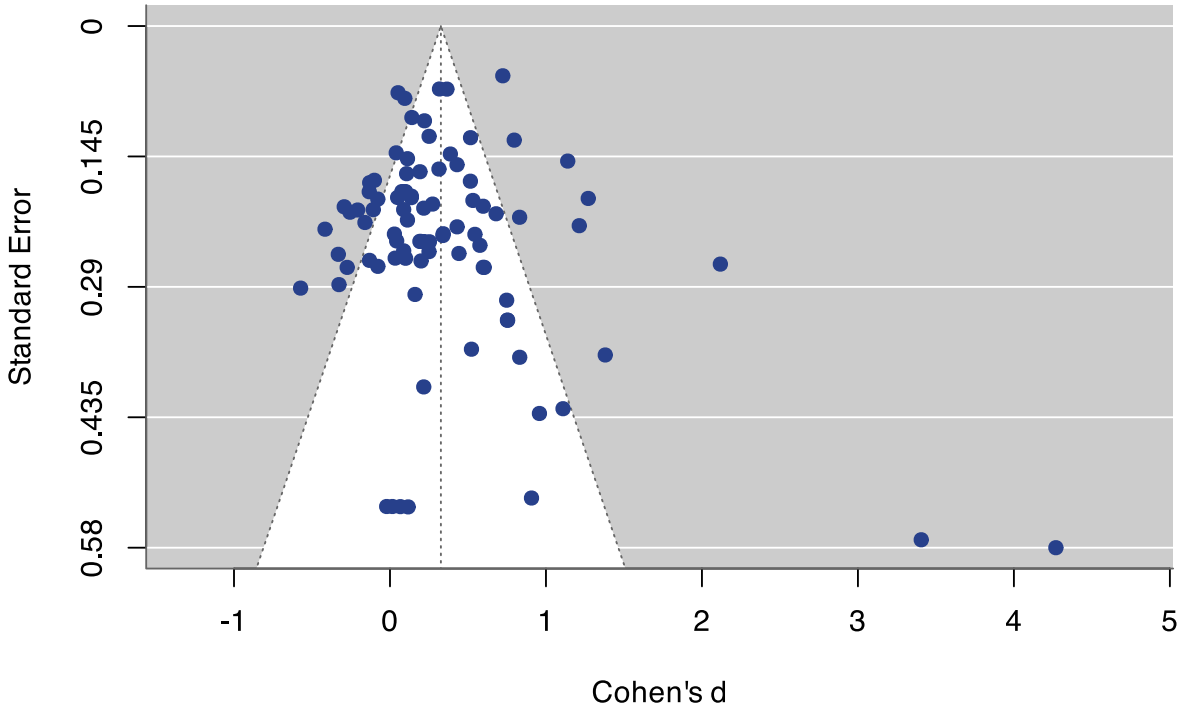


Figure 4

Radial (Galbraith) plot for assessment of heterogeneity.

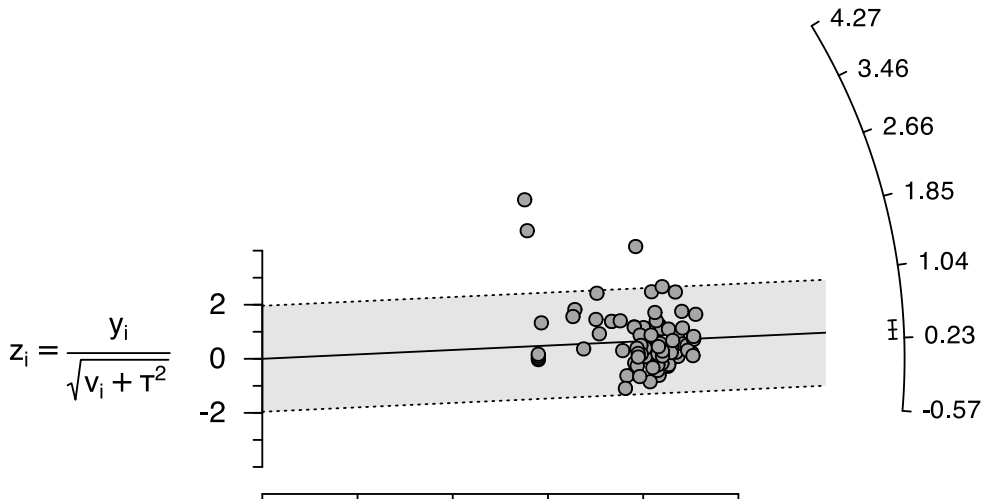
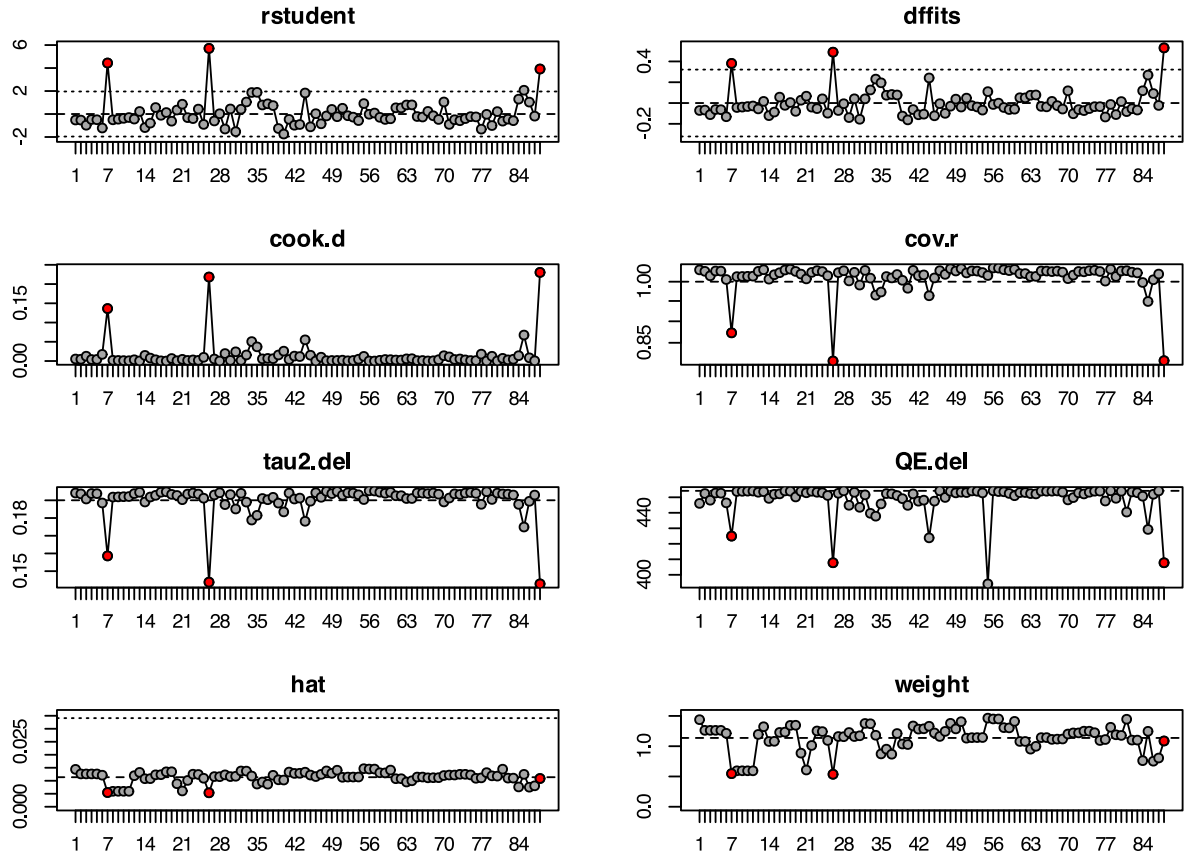


Figure 3

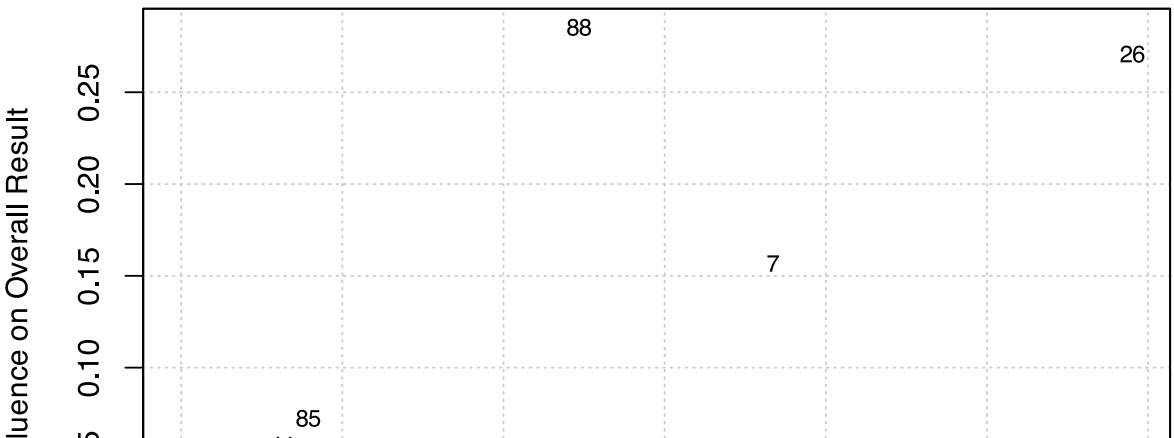
Influence diagnostics: Cook's distances and DFFITS for each included effect size.



Moderator Analyses

Figure 5

Baujat plot: contribution of each study to overall heterogeneity (x-axis) and influence on the pooled estimate (y-axis).



Magnitude and Interpretation of the Overall Effect

The estimated effect size falls in the small-to-medium range by conventional benchmarks (Cohen, 1988). This is broadly consistent with findings from related domains: Anderson et al. (2010) reported effects in the range of $d = 0.24–0.40$ for violent game effects on aggression; Greitemeyer and Mügge (2014) found an overall effect of $d = 0.27$ for prosocial content on prosocial outcomes. Our estimate accounts for the nested structure of the data and corrects for scale direction across heterogeneous outcome measures—methodological improvements over earlier work.

The substantial heterogeneity ($I^2_{\text{total}} \approx 80\%$) is not surprising given the diversity of game types, study designs, outcome measures, and participant populations included. Rather than treating heterogeneity as a problem, we interpret it as a call for moderator analysis.

Moderators: Content vs. Context

A key distinction in this meta-analysis was between game *content* (prosocial by design or narrative) and game *context* (cooperative play mode). Moderator analyses indicated that neither game content (prosocial vs. neutral/violent: $b = -0.50$, SE = 0.31, $z = -1.60$, $p = .110$) nor game context (cooperative vs. competitive/solo: $b = -0.15$, SE = 0.31, $z = -0.48$, $p = .632$) was a statistically significant predictor of effect size. Although the point estimates suggest a trend toward smaller effects in explicitly prosocial-content games, the confidence intervals are wide and the evidence remains inconclusive given the current sample size. This finding aligns with recent experimental evidence. Shoshani and Krauskopf (2021) showed that cooperative violent games produced more prosocial behavior than cooperative neutral games, suggesting an additive or synergistic benefit of combining prosocial content with cooperative context. Zhang et al. (2025) similarly demonstrated that cooperative game settings promoted sharing behavior, particularly among preschool girls. Sherrick et al. (2025) found that cooperative game modes strengthen prosocial injunctive norms even in contexts otherwise associated with antisocial online behavior.

Taken together, these results support a nuanced picture: both what a game is *about* and how it is *played* independently contribute to prosocial outcomes, and these factors may interact.

The Role of Study Design

Experimental and correlational studies may yield different effect estimates due to self-selection into game types, demand characteristics, and the distinction between immediate and habitual effects. The moderator analysis examining study design as a predictor found no significant effect ($b = -0.13$, SE = 0.22, $z = -0.61$, $p = .543$), indicating that experimental and correlational studies did not differ systematically in their pooled effect sizes within this dataset. This mirrors the pattern observed in the violent-game literature, where experimental studies typically yield smaller but more internally valid estimates (Anderson et al., 2010). Participant age also did not significantly moderate the effect (b = -0.0017, p = 0.89), suggesting that the prosocial/cooperative game effect is relatively stable across age groups represented in the literature.

LLM-Assisted Effect-Direction Coding

A methodological contribution of this meta-analysis is the use of an LLM to assist in coding the direction of effect for each outcome measure. This approach proved efficient for handling the semantic heterogeneity of prosocial outcome measures, which ranged from behavioral helping tasks to self-report empathy scales to aggression measures requiring sign reversal. The LLM assigned unambiguous codes for 94.3% of effect sizes, with the remaining cases resolved through human consensus. This hybrid human-LLM coding strategy is consistent with emerging evidence that LLMs can approximate human coder accuracy in meta-analytic tasks at scale (Blaizot et al., 2022).

Publication Bias

The funnel plot inspection and formal tests suggested asymmetry consistent with publication bias (Egger's test: $z = 3.36$, $p < .001$; Kendall's $\tau = 0.156$, $p = 0.031$). This is a common finding in the social psychology literature and suggests that the true effect may be somewhat smaller than the estimated pooled effect (Simonsohn et al., 2014).

Limitations

Several limitations should be noted. First, the literature search was conducted using Google Scholar as the primary database, which—while broad—may not have captured all eligible studies. A more comprehensive search using PsycINFO, Web of Science, and CENTRAL is recommended for future updates. Second, the range of outcome measures was wide, and despite LLM-assisted direction coding, some degree of construct heterogeneity remains. Third, most studies were conducted in WEIRD (Western, Educated, Industrialized, Rich, Democratic) populations, limiting generalizability. Fourth, the moderator analyses should be treated as exploratory given the relatively modest number of independent studies.

Conclusions

This meta-analysis provides evidence that playing games with prosocial content or in cooperative contexts produces positive social outcomes, including greater helping behavior, empathy, and cooperation. The effect is moderate in size and highly heterogeneous. Although neither game content nor game context emerged as statistically significant moderators in the current sample, the theoretical distinction between *what* a game is about and *how* it is played remains meaningful and warrants continued investigation with larger primary datasets. These findings are broadly consistent with the General Learning Model's prediction that game content shapes social cognition and behavior, while also highlighting the potential role of cooperative play structure as a social pathway. Practitioners and game designers can draw cautious optimism from these findings: embedding cooperative mechanics and prosocial narratives into educational and serious games appears associated with positive social outcomes, though effect sizes vary considerably across contexts.

Future research should employ pre-registered designs, standardized outcome batteries, and longitudinal follow-ups to establish the persistence of game-based prosocial effects beyond the lab. The use of LLM assistance in meta-analytic workflows—particularly for effect-direction coding and abstract screening—merits further validation and methodological formalization.

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Table 2

Overview of studies included in the meta-analysis (first 20 shown).

Study	Year	Design	Game type	Prosocial	Cooperative	n effects	N
Creighton_2014	2014	eksperyment	video	0	1	2	32
Eden_2014	2014	eksperyment	video	0	1	2	112
Middleton_2014	2014	korelacje	video	1	0	1	136
Vang_2014	2014	eksperyment	video	0	1	1	99
Alladin_2015	2015	korelacje	video	0	1	1	695
Beene_2015	2015	eksperyment	video	0	1	4	14
Happ_2015	2015	eksperyment	video	1	0	2	80
Kaufman_2015	2015	eksperyment	inne	1	NA	3	193
Lim_2015	2015	korelacje	video	1	0	1	97
Liu_2015	2015	eksperyment	video	1	0	2	49
Smith_2015	2015	korelacje	video	1	0	1	387
Andrews_2016	2016	eksperyment	video	1	0	1	94
Badatala_2016	2016	eksperyment	video	0	1	1	30
Coyne_2016	2016	korelacje	video	1	1	2	201
Lobel_2016	2016	korelacje	video	0	1	1	184
Loparev_2016	2016	eksperyment	video	0	1	1	132
Najarian_2016	2016	eksperyment	video	1	0	1	97
Roussos_2016	2016	eksperyment	video	1	0	1	165
Stiff_2016	2016	eksperyment	video	0	1	4	58
Velez_2016	2016	eksperyment	video	0	1	2	85

Table 4

Distribution of LLM-assigned (ChatGPT) effect direction codes.

LLM code	n	Interpretation	%
+	70	Higher score = more prosocial (retain sign)	79.5
-	13	Higher score = less prosocial (reverse sign)	14.8
?	5	Ambiguous – resolved manually	5.7

Table 9

Meta-regression results: moderators of the prosocial game effect.

Moderator	b	SE	z	p
Game content (prosocial)	-0.348	0.259	-1.34	0.179
Game context (cooperative)	-0.133	0.259	-0.52	0.606
Study design (experimental)	-0.027	0.160	-0.17	0.865